FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2019

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. CD

Sol. Take torque of tension about IAOR it will first clockwise and then anticlockwise torque so it will come to rest after covering certain distance. When spool at rest at extreme right block will also be remain at rest but at same initial position (according to energy conservation)

2. BD

Sol. Current disc to rotated $dI = \dfrac{\sigma 2\pi r dr}{\dfrac{2\pi}{\omega}} = \sigma\omega r dr$

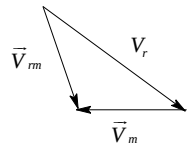
$$B = \sum_0 \frac{\mu_0 \sigma \omega r dr}{2r} = \frac{\mu_0 \omega}{2}\left[A - B\frac{1}{2}\right]$$

$$M = \sum_0^1 (\sigma\omega r dr)\pi r^2$$

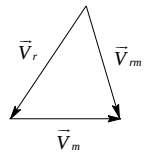
$$M = \omega\pi\left[\frac{A}{4} - \frac{B}{5}\right]$$

3. ABD

Sol. If man is moving in positive x direction



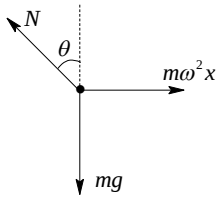
If man is moving in negative x direction



4. B

Sol. $\tan\theta = \dfrac{\omega^2 x}{g}$

$\dfrac{dy}{dx} = \dfrac{\omega^2 x}{g} \Rightarrow y = \dfrac{\omega^2 x^2}{2g}$



5. B

Sol. Optical path length of ray entering through S_1

$x_1 = \int_0^L \mu dx = L + \dfrac{aL^2}{2}$

Path difference of ray from S_1 and $S_2 = \Delta x = \dfrac{aL^2}{2}$

For minima at O

$\dfrac{aL^2}{2} = \dfrac{\lambda}{2}$

6. D

Sol. Energy of nth state in H-atom is same as energy if 3nth state in Li^{++}.

7. A

Sol. (P) X is a resistor

$m\dfrac{dv}{dt} = -1\ell B = \dfrac{-B^2\ell^2 v}{R}$

$v = v_0 e^{-t/T}\ T = \dfrac{mR}{B^2\ell^2}$

(Q) X is an inductor

$x = \dfrac{v_0}{\omega}\sin\omega t, \omega = \dfrac{B\ell}{\sqrt{mL}}$

$x = \dfrac{v_0}{\omega}\sin\omega t, \omega = \dfrac{B\ell}{\sqrt{mL}}$

(R) $\theta = CV$

$F = m\dfrac{dv}{dt} = F_0 - B^2e^2C\dfrac{dv}{dt}$

$a = \dfrac{F_0}{m + B^2e^2C}$

8. A

Sol. Initial charge on the capacitor $q_0 = CV_0 = 75\times10^{-3}\times\dfrac{640}{3} = 16C$

The charged on the capacitor decays as

$$q = q_0 e^{-t/RC}$$

at $t = 2.5\ln(4)$ minutes = 150 ln(4) s

$$q = 16 \times e^{-\ln(4)} = 4C \quad \because RC = 150 \text{ s}$$

Total heat dissipated in the resistor in the given time = $\frac{q_0^2 - q^2}{2C} = 1.6kJ$ = heat imparted to the gas

1 Work done by the gas at constant pressure = $P\Delta V = nR\Delta T \approx 0.60kJ$

2 Increment in the internal energy $\Delta U = Q - W = 1.6 - 0.6 = 1kJ$

3 $\gamma = 1 + \frac{nR\Delta T}{\Delta U} = 1.6$

9. D
10. C

SECTION – D

11. 00002.99

12. 00001.67

13. 00000.25

14. 00001.00

Sol. The change in internal energy of the gas is DU = 1.5 nR (DT)= 1.5 (1 mol) (253J/mol-K) (2K)
= 25 J.
The heat given to the gas = 42 J
The work done by the gas is DW = DQ – DU= 42 J – 25 J = 17 J.
If the distance moved by the piston is x, the work done is DW = (100 kPa) (8.5 cm^2) x.
Thus, (105 N/m^2) (8.5 × 10–4 m^2) x = 17 J or, x = 0.2m = 20 cm.

15. 00002.00

Sol. $t_1 = RC$; $t_2 = R/L$

→LC = t_1t_2 = 0.1 sec -> T = $2\pi\sqrt{\frac{1}{LC}}$

T = 2 sec.

16. 00000.73

Sol. Impulse due to 'N' would act at the centre of the colliding parts
Applying Impulse momentum theorem

$$-\int N.dt = M\left(v - v_0\right) \qquad ---(1)$$

& Applying Angular impulse, angular momentum theorem

$$\int N\left(\frac{L-x}{2}\right).dt = \frac{ML^2}{12}\omega \qquad ---(2)$$

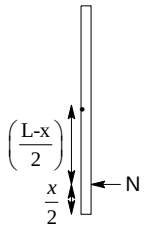
Also $e = \frac{\omega\left(\frac{L-x}{2}\right) - v}{v_0} \qquad ---(3)$

From (1), (2) & (3)

$$v\left[1+3\left(\frac{L-x}{2}\right)^2\right]=v_0\left[3\frac{(L-x)^2}{L^2}-e\right]$$

$\therefore$ *for* $v>0$

$$x<L\left[1-\sqrt{\frac{e}{3}}\right]$$



For inequality to hold true for all 'e'

$$x<L\left[1-\frac{1}{\sqrt{3}}\right]$$

$$or\ x<\left(\sqrt{3}-1\right),\ \therefore x_{\max}=0.732$$

17. 00008.89

Sol. Power absorbed by shell

$$\Rightarrow \frac{\mathrm{P}\times 2\pi\left(1-\cos 37\right)}{4\pi}$$

$$\Rightarrow \frac{\mathrm{P}}{10}$$

For equilibrium

$$\Rightarrow \frac{P}{10}=3\pi R^2\sigma T^4$$

$$\therefore T^4=\frac{P}{30\pi R^2\sigma}=\frac{2400\sigma\pi}{30\pi\times 9\times\sigma}\Rightarrow 8.89$$

18. 00005.66

Sol. Applying conservation of angular momentum w. r. t. the centre of earth,

$$4R\times mv_0\sin 30=mvR, v=2v_0$$

Applying energy conservation

$$\frac{1}{2}m\left(2v_0\right)^2-\frac{GMm}{R}=-\frac{GMm}{4R}+\frac{1}{2}mv_0^2$$

$$\therefore v_0\Rightarrow\sqrt{\frac{GM}{2R}}\Rightarrow\sqrt{\frac{64\times 10^6}{2}}=4\sqrt{2}\times 1000 m/s$$

$$\therefore x=4\sqrt{2}\Rightarrow 5.66$$

Chemistry

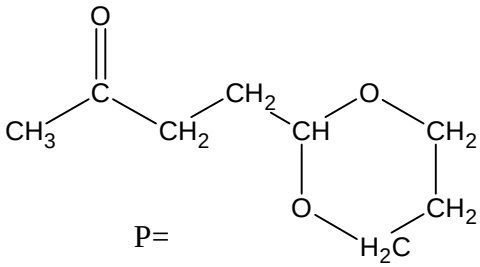
PART – II

SECTION – A

19. AB

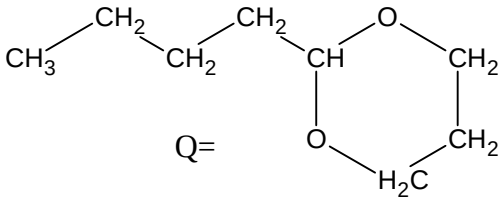
Sol.

(A)



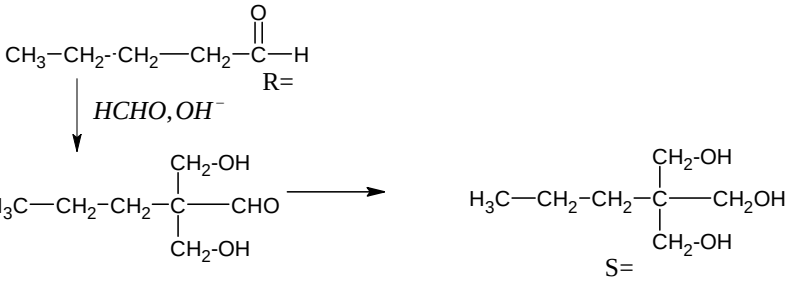
P=

(B)



Q=

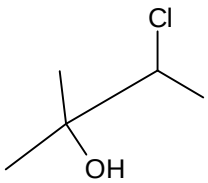
(C)



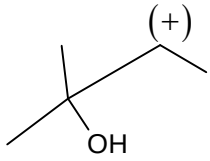
20. CD

Sol.

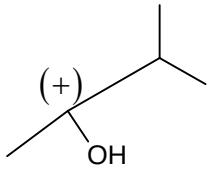
R=



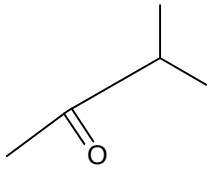
Intermediate I=



Intermediate II=



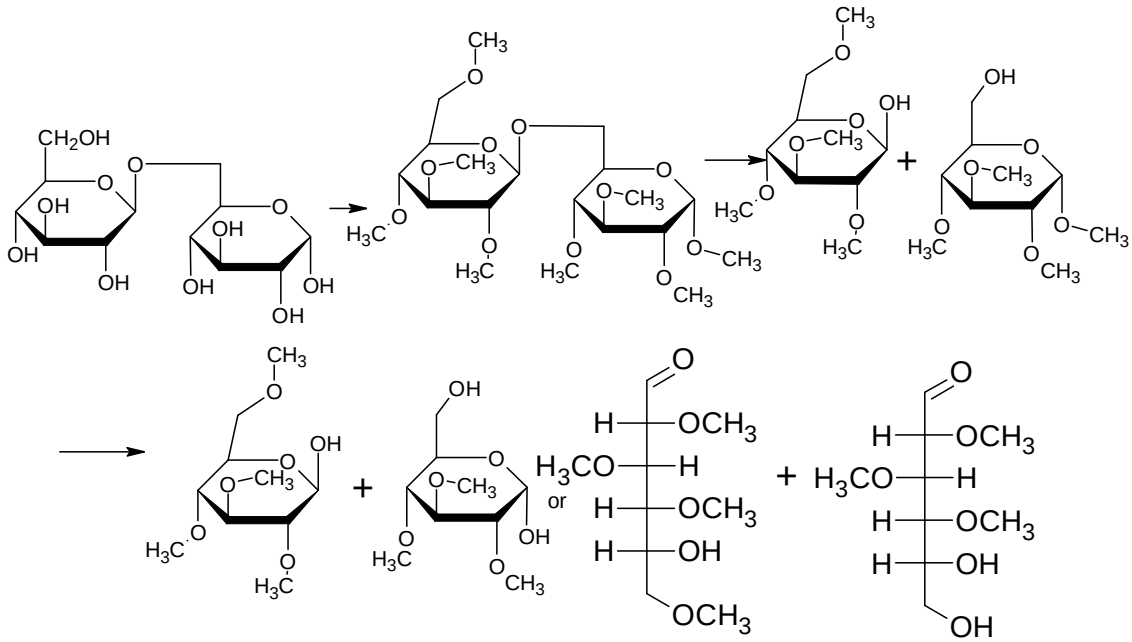
T=



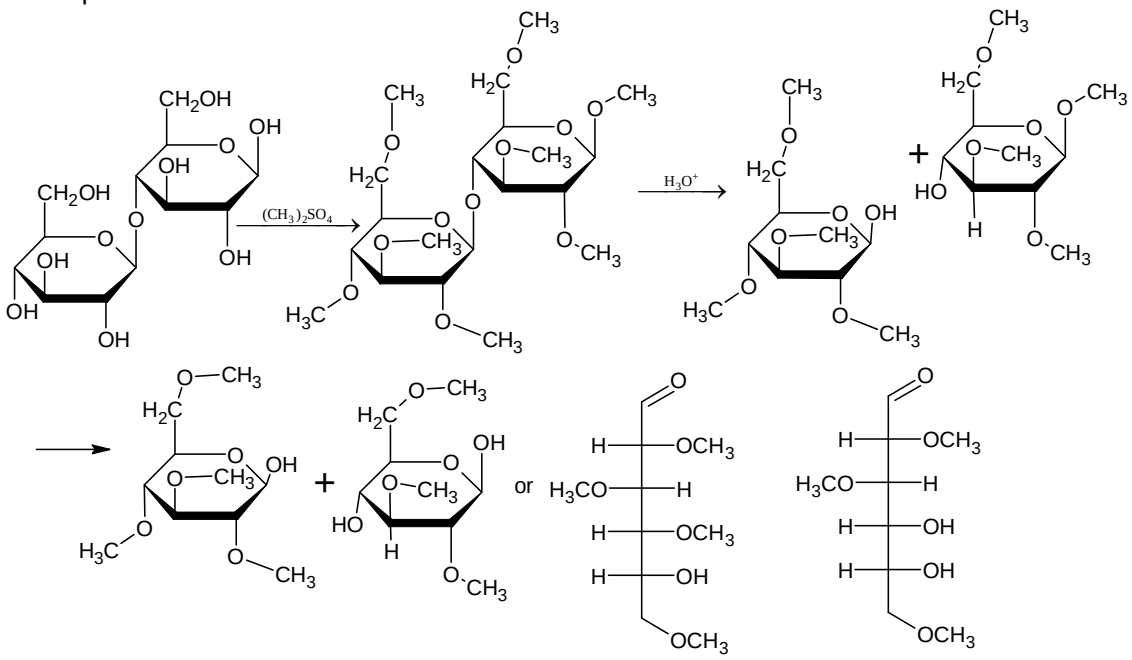
21. ABC

Sol. For A and D there is no $\alpha-$linkage.

For B part



For D part



22. BCD

Sol. In Lyophilic sols, dispersion medium has affinity with dispersed phase.

23. ABC

Sol. 0.005mole $NaOH$+0.00235mole Na_2CO_3 (0.25/106)

so H+ required = 0.0073 mole

$0.00735 = x \times 15 \times 10^{-3}$

$x = 0.49$

0.5gm (0.0059 mol) $NaHCO_3$ is added. so total mole of $NaHCO_3$ =0.00825

$0.00825 = y \times 25.5 \times 10^{-3}$

$y = 0.323$

$\frac{x}{y} = 1.5$

$0.00736 = 0.32 \times V(L)$

$y = 23ml$

0.00235*mole* of $NaHCO_3$ (already present)+0.02976 mole ($\frac{2.5}{84}$ mole) of $NaHCO_3$ is added so toatal $NaHCO_3$ = 0.03211 mol $NaHCO_3$

$0.03211 = 0.49 \times V \times 10^{-3}$

$V = 65.5ml$

24. AC

Sol. C_4H_{10}, CH_3CH_2OH, NH_3 and I_2 have high BP so low vapour pressure.

25. A

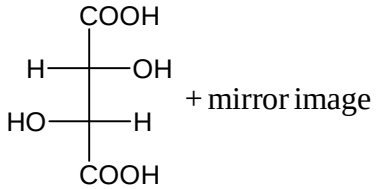
Sol. $ZnS \longrightarrow ZnO\,(no\ reduction)$

Zn *extraction does not involve self reduction*

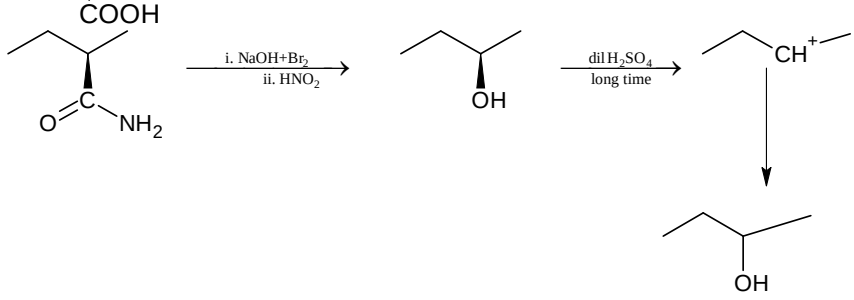
$Cu_2S \longrightarrow Cu_2O$ is not calcination.

26. A

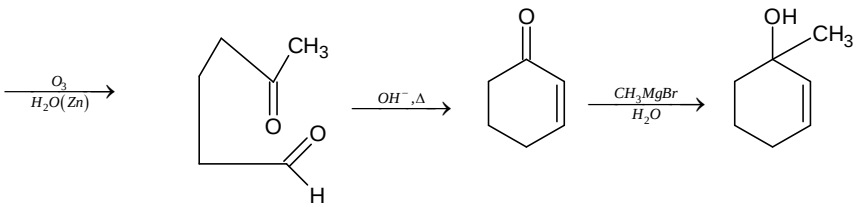
Sol.

(P) 

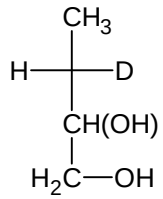
+ mirror image

(Q) 

$(\pm)$

(R) 

(S)



27. A

Sol. (P) $\rightarrow$(2), (Q) $\rightarrow$(1), (R) $\rightarrow$(5), (S) $\rightarrow$(3)

HCN is a weak acid so maximum pH at end point, CH_3COOH is a stronger acid than HCN, Strong base is CN^-, $K_b = 10^{-4}$, weak base is CH_3COO^- as it's $K_b = 5\times10^{-10}$

28. A

Sol. (P) $\rightarrow$(1, 5), (Q) $\rightarrow$(1), (R) $\rightarrow$(3), (S) $\rightarrow$(4)

(1) $Sn(NO_3)_2 \equiv Sn^{2+}, NO_3^-$

Sn^{+2} can further oxidize into Sn^{+4} and It is basic radical NO_3^- can't further oxidize.

(2) $KNO_3 \equiv K^+, NO_2^-$

K^+ can't be further oxidize but NO_2^- can be oxidize into NO_3^-

(3) $FeC_2O_4 \equiv Fe^{2+}, CO_4^{2-}$

Fe^+ can be oxidize into CO_2 so both radical

(4) $NaCO_3 \equiv Na^+, CO_3^{2-}$

both ions can't be oxidized further so no radical

(5) $BaSO_3 \equiv Ba^{2+}, SO_3^{2-}$

Ba^{2+} can't be oxidized but SO_3^{2-} can be oxidized into SO_4^{2-} only acid radical.

SECTION – D

29. 00004.00

Sol. $2A + B \longrightarrow C + 2D$

$x \quad 2x \quad 0$

$(x-\frac{x}{2}) \quad (2x-\frac{x}{4}) \quad \frac{x}{4}$

$\frac{x}{2} \quad \frac{7x}{4} \quad \frac{x}{4}$

Rate of reaction = $k(\frac{x}{2})(\frac{7x}{4})^2 = 49kx^3/32$

Rate of decomposition of $A = 2\times49kx^3/32 = 49kx^3/16$

30. 00000.11

Sol. $V_2O_5 + 10H^+ + 3Zn \longrightarrow 2V^{2+} + 5H_2O + 3Zn^{2+}$

$2V^{2+} + 2I_2 + 2H_2O \longrightarrow 4I^- + 2VO^{2+} + 4H^+$

Add

$V_2O_5 + 6H^+ + 3Zn + 2I_2 \longrightarrow 3Zn^{2+} + 3H_2O + 2VO^{2+} + 4I^-$

10 gm V_2O_5=(10/ 180) mol

so moles of I_2 required=$2\times\frac{10}{180}=\frac{1}{9}=0.11$

31. 00099.83

Sol.

For precipitation of AgI, I^- requires=$\frac{8.5\times10^{-17}}{0.1}=8.5\times10^{-16}$

For precipitation of Hg_2I_2, I^- requires=$\sqrt{\frac{2.5\times10^{-26}}{0.1}}=5\times10^{-13}$

so first AgI will precipitate.

Maximum conc. of I^- at which only AgI will precipitate=5×10^{-13}

Now $[Ag^+]=\frac{8.5\times10^{-17}}{5\times10^{-13}}=1.7\times10^{-4}$

so % precipitation= $\frac{0.1-1.7\times10^{-4}}{0.1}\times100=(1-17\times10^{-4})\times100=(1-17\times10^{-4})\times100=99.83$

32. 00002.97

Sol. By Henry Law

$p=\chi\times K_H--------(i)$

$\frac{5}{9}\text{molal}\Rightarrow\chi=\frac{5/9}{5/9+1000/18}=\frac{10}{1010}$

Put in (i)

$p=\frac{10}{1010}\times300=\frac{300}{101}=2.970atm$

33. 00001.00

Sol. $Ag^+ + e\longrightarrow Ag$

$E_{cell}=0.5=0.8+\frac{0.06}{1}\log[Ag^+]$

$[Ag^+]=10^{-5}\text{mol/litr}=\frac{\text{mass of Ag}/108}{0.5}$

mass of Ag=$54\times10^{-5}g$

% of silver=$\frac{54\times10^{-5}}{54\times10^{-3}}\times100=1$

34. 00021.00

Sol. $E_{\text{isotpic field}}=PE$

$E_{\text{Ligand field}}=6\times(-0.4\Delta_0)+3\times PE$

$CFSE=2\times PE+6\times(-0.4\Delta_0)$

$-8800=2\times20800-2.4\times\Delta_0$

$\Delta_0=21000$

$x=21.000$

35. 00000.54

Sol. $\ln\frac{P_2}{P_1}=\frac{\Delta H}{R}\left[\frac{1}{T_1}-\frac{1}{T_2}\right]$

$\ln\frac{P_2}{760}=\frac{9720}{2}\left[\frac{1}{373}-\frac{1}{348}\right]$

P_2 =298torr

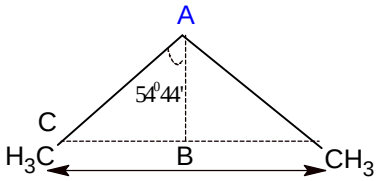
$\frac{298-P_{sol.}}{298}=\frac{0.1}{0.1+\frac{1000}{18}}$

P_{sol} =297.46torr

Lowering in VP=(298-297.46)=0.54torr

36. 00002.40

Sol.



$\text{Sin}(54^0 44')=\frac{CB}{AC}$

Terminal carbon carbon distance= $2\times CB$

$=2\times\ 1.5\ \times\text{Sin}(54^0 44')$

$=2\times\ 1.5\ \times 0.8\ A^0$

$=2.4\,\text{Å}$

Mathematics

PART – III

SECTION – A

37. BD

Sol. $f\left(\frac{1}{x}\right)=\int_1^{1/x}\frac{\ln t}{(1+t)}dt=\int_1^{x}\frac{-\ln u}{\frac{1+u}{u}}\times\frac{-1}{u^2}du=\int_1^{x}\frac{\ln t}{t(1+t)}dt$

Now, $f(x)+f\left(\frac{1}{x}\right)=\int_1^{x}\frac{\ln t}{(1+t)}\left(1+\frac{1}{t}\right)dt=\int_1^{x}\frac{\ln t}{t}dt=\left[\frac{(\ln t)^2}{2}\right]_1^{x}=\frac{(\ln x)^2}{2}$.

38. ABCD

Sol. $2\sin\left(\frac{x}{2}\right)(\cos^2 x-\sin^2 x)=\cos^2 x-\sin^2 x$

$\left(2\sin\left(\frac{x}{2}\right)-1\right)\cos 2x=0$

$\cos 2x = 0$ or $\sin\frac{x}{2}=\frac{1}{2}$

$2x = n\pi+\frac{\pi}{2}$

$\frac{x}{2}=k\pi+(-1)^k\frac{\pi}{6}$ $\quad (n, k\in I)$

$x=\frac{n\pi}{2}+\frac{\pi}{4}$

or $x = 2k\pi+(-1)^k\frac{\pi}{3}$

$x=\frac{n\pi}{2}+\frac{\pi}{4}$, then $\sin 2x = \pm 1$ and if $x = 2k\pi+(-1)^k\frac{\pi}{3}$

$\cos x=\cos\frac{\pi}{3}=\frac{1}{2}$ and $\cos 2x=\cos\frac{2\pi}{3}=-\frac{1}{2}$.

39. AD

Sol. $\left|\frac{\frac{c}{3}}{\sqrt{\frac{1}{3^2}+\frac{1}{4^2}}}\right|=c \Rightarrow c = 6, 1.$

40. CD

Sol. $\sin^{-1}x>\frac{\pi}{2}-\sin^{-1}x$

$\sin^{-1}x>\frac{\pi}{4}$

$x>\sin\frac{\pi}{4}\Rightarrow x>\frac{1}{\sqrt{2}}$ and $x\in(0, 1)$

$x\in\left(\frac{1}{\sqrt{2}}, 1\right)$.

41. ABD

Sol. $2\cot^{-1}7 = 2\tan^{-1}\frac{1}{7} = \cos^{-1}\left(\frac{1-\frac{1}{49}}{1+\frac{1}{49}}\right) = \cos^{-1}\left(\frac{24}{25}\right)$

$2\cot^{-1}7 + \cos^{-1}\frac{3}{5} = \cos^{-1}\frac{24}{25} + \cos^{-1}\frac{3}{5} = \cos^{-1}\left(\frac{44}{125}\right)$

since $0 < \cos^{-1}\left(\frac{44}{125}\right) < \frac{\pi}{2}$

$\cos\theta = \frac{44}{125}$, $\operatorname{cosec}\theta = \frac{125}{177}$

$\cot\theta = \frac{44}{117}$.

42. ABC

Sol. $1+\cot\theta-\cot\frac{\theta}{2} = 1 + \frac{\cot^2\frac{\theta}{2}-1}{2\cot\frac{\theta}{2}} - \cot\frac{\theta}{2} = \frac{-\cot^2\frac{\theta}{2}-1+2\cot\frac{\theta}{2}}{2\cot\frac{\theta}{2}} = \frac{-\left(\cot\frac{\theta}{2}-1\right)^2}{2\cot\frac{\theta}{2}} \le 0$

$\cot\frac{\theta}{2} > 0$ in $0 < \frac{\theta}{2} < \frac{\pi}{2}$

$\cot\frac{\theta}{2} > 0$ in $0 < \theta < \pi$.

$\therefore\ \theta \in (0, \pi)$.

43. A

Sol. (P). $\frac{\log_2(x+3)}{x^2+8x+15}$

$x + 3 > 0 \Rightarrow x > -3 \quad \therefore x \in (-3, \infty)$

Also, $x^2 + 8x + 15 \neq 0$ or $(x + 3)(x + 5) \neq 0$

$X \neq -3, -5$.

$\therefore$ domain is $(-3, \infty)$.

(Q). Domain of $\sqrt{1-\log_3 x}$

$1 - \log_3 x \ge 0 \Rightarrow \log_3 x \le 1$

$x > 0$ and $\log_3 x \le \log_3 3$

$\Rightarrow x > 0$ and $x \le 3$

$\Rightarrow x \in (0, 3]$.

(R). Domain of $\frac{\sqrt{4-x^2}}{\sin(2-x)}$

$4 - x^2 \ge 0 \Rightarrow x \in [-2, 2] = D_1$

$-1 \le 2 - x \le 1$ or $x \le 3$ or $x \ge 1$

$\therefore x \in [1, 3] = D_2$

But $\sin(2 - x) \neq 0$

$\Rightarrow x \neq 2$

$\therefore$ domain is $D_1 \cap D_2 - \{2\}$

$= [1, 2)$.

(S). Domain of $\log[1 - \log_9(x^2 - 3x + 9)]$

$1 - \log_9(x^2 - 3x + 9) > 0$

$\log_9(x^2 - 3x + 9) < \log_9 9$
$x^2 - 3x + 9 < 9$
$x^2 - 3x < 0$
$x \in (0, 3)$.

44. A

(P). $s = (a + b - c)(a - b + c)$
$s^2 = (2(s - b)\, 2(s - c))^2$
$s(s - a)(s - b)(s - c) = 16(s - b)^2 (s - c)^2$

$$\frac{(s-b)(s-c)}{s(s-a)} = \frac{1}{16}$$

$$\tan^2\frac{A}{2} = \frac{1}{16} \text{ or } \tan\frac{A}{2} = \frac{1}{4}$$

$$\tan A = \frac{2\tan\frac{A}{2}}{1-\tan^2\frac{A}{2}} = \frac{8}{15}$$

$$15\tan A = \frac{8}{15}\times 15 = 8.$$

(Q). $\Delta = \frac{1}{2}ab\sin c = 9\sin c$

$$\cos(A-B) = \frac{4}{5} \Rightarrow \frac{1-t^2}{1+t^2} = \frac{4}{5}$$

$\frac{2t^2}{2} = \frac{1}{9}$ by CD $\therefore$ $t = \tan\frac{A-B}{2} = \frac{1}{3}$

$$\frac{a-b}{a+b}\cot\frac{c}{2} = \frac{1}{3} \quad \therefore \quad \frac{3}{9}\cot\frac{c}{2} = \frac{1}{3} \Rightarrow \cot\frac{c}{2} = 1$$

$\frac{c}{2} = 45 \Rightarrow c = 90°$

Δ 9 sq. units.

(R). In Δ $a + b > c$ and in quadrilateral $a + b + c > d$
$(a - b)^2 + (b - c)^2 + (c - a)^2 > 0$
$2(a^2 + b^2 + c^2) \geq 2(ab + bc + ca)$
add $a^2 + b^2 + c^2$
$3(a^2 + b^2 + c^2) \geq a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
$\Rightarrow (a + b + c)^2 > d^2$

$$\therefore \frac{a^2+b^2+c^2}{d^2} \geq \frac{1}{3}$$

= minimum value is $\frac{1}{3}$.

(S). PR $\perp$ QR $\Rightarrow m_1 m_2 = -1$
$3x + 4y = 13$

$\Delta PQR = 7 \Rightarrow \frac{1}{2}(9 - 4x + 3y) = \pm 7$

$4x - 3y + 5 = 0$ and $4x - 3y - 23 = 0$
There will be two points by solving above equation we will get.

45. C

Sol. (P). Put, 5x = t in $\int_0^\infty \frac{\sin 5x}{x}dx$ we get,

$\int_0^\infty \frac{\sin t}{t}dt = \frac{\pi}{2}$.

(Q). $I = \frac{1}{2}\int_0^\infty \frac{\sin(a+b)x + \sin(a-b)x}{x}dx$

If a > b, then $I = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$

If a < b, then $I = \frac{\pi}{4} - \frac{\pi}{4} = 0$.

(R). $\int_0^\infty \frac{\sin^2 x}{x^2}dx = \left|\sin^2\left(-\frac{1}{x}\right)\right|_0^\infty - \int_0^\infty \frac{2\sin x\cos x}{x}dx = \frac{\pi}{2}$.

(S). $\int_0^\infty \frac{\sin^3 x}{x}dx = \int_0^\infty \left(\frac{3\sin x - \sin 3x}{4x}\right)dx = \frac{3}{4}\int_0^\infty \frac{\sin x}{x}dx - \frac{1}{4}\int_0^\infty \frac{\sin 3x}{x}dx$

$= \frac{3}{4}\times\frac{\pi}{2} - \frac{1}{4}\times\frac{\pi}{2} = \frac{\pi}{4}$.

46. D

Sol. (P). I l, m, n be the d.r. of the line, then

l – cm – bn = 0, – cl + m – an = 0, – bl – am + n = 0

Eliminating l, m, n we get

$\begin{vmatrix} 1 & -c & -b \\ -c & 1 & -a \\ -b & -a & 1 \end{vmatrix} = 0 \Rightarrow a^2 + b^2 + c^2 + 2abc = 1.$

(Q). The equation of the lines in the symmetrical form is $\frac{x-1}{a} = \frac{y-0}{1} = \frac{z-2}{b}$

and $\frac{x-3}{c} = \frac{y-0}{1} = \frac{z-4}{d}$.

The lines are perpendicular if a . c + 1 . 1 + b . d = 0.

(R). The equation of plane ABC is $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 3$ and (a, b, c) lies on it so, $\frac{\alpha}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} = 3$.

(S). If – 1 < x < 0, then $-1 \le \sin\pi x < 0 \Rightarrow 0 < x\sin\pi x < 1 \Rightarrow [x\sin\pi] = 0$.

If 0 < x < 1, then $0 < \sin\pi x < 1 \Rightarrow 0 < x\sin\pi x < 1 \Rightarrow [x\sin\pi x] = 0$

$\therefore$ f(x) = 0 $\forall$ x $\in$ (– 1, 1), so continuous and differentiable but

$\lim_{x\to 1^+} f(x) = -1$ and $\lim_{x\to -1^-} f(x) = -1$

$\therefore$ f(x) is not continuous at x = 1 and – 1 and hence not differentiable at x = – 1, 1.

SECTION – D

47. 00002.00

Sol. Equation of the circle is $(x - 1)2 + (y - 1)^2 + k(x + y - 2) = 0$

As the circle passes through (2, 2) $\Rightarrow$ x = –1.

Equation $x^2 + y^2 - 3y + 4 = 0$

Radius = $\sqrt{\frac{9}{4}+\frac{9}{4}-4}=\frac{1}{\sqrt{2}}$

(diameter)$^2=\left(\sqrt{2}\right)^2=2$.

48. 00216.00

Sol. $\sec A\sec B\sec C(1+\sec A)(1+\sec B)(1+\sec C)=\frac{\tan^2 A\tan^2 B\tan^2 C}{8\sin^2\frac{A}{2}\sin^2\frac{B}{2}\sin^2\frac{C}{2}}$

or $\tan A\tan B\tan C\geq 3\sqrt{3}$ and $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\leq\frac{1}{8}$

$\Rightarrow$ minimum value = 216.

49. 00015.00

Sol. $\frac{z_2}{z_1}=\frac{OQ}{OP}e^{i\theta}$

$\frac{z_3}{z_1}=\frac{OR}{OP}e^{i2\theta}$

$z_2^2=z_1^2\cos^2\theta\times\frac{z_3}{z_1\cos^2\theta}\Rightarrow\frac{\cos^2\theta}{\cos 2\theta}=\frac{2+\sqrt{3}}{2\sqrt{3}}$

$\Rightarrow\cos^2\theta=\frac{2+\sqrt{3}}{4}\Rightarrow 2\cos^2\theta=\frac{2+\sqrt{3}}{4}$

$\Rightarrow 2\cos^2\theta=1+\cos 30^\circ\Rightarrow\theta=15$

50. 00002.00

Sol. $\log_{\sqrt{5}}\tan\theta\sqrt{\frac{3}{\log_{\sqrt{5}}\tan\theta}+3}=-\sqrt{6}$

$\Rightarrow x\sqrt{\frac{3}{x}+3}=-\sqrt{6}$ ... (1) $\left(x=\log_{\sqrt{5}}\tan\theta\right)$

$\Rightarrow\frac{x\sqrt{x+1}}{\sqrt{x}}=-\sqrt{2}$

$\Rightarrow x(x^2+x-2)=0$

$\therefore x=0,1,-2$

but $x<0$ (from (1))

$\therefore x=-2$

$\log_{\sqrt{5}}\tan\theta=-2$

$\therefore$ two solutions.

51. 00000.41

Sol. $I=\int_0^1 0dx+\int_1^{\sqrt{2}}dx+\int_{\sqrt{2}}^{1.5}2dx$

$=\left(\sqrt{2}-1\right)+2\left(1.5-\sqrt{2}\right)$

$=2-\sqrt{2}$

52. 00052.16

Sol. $|z|^2 + (3 - 4i)z + (3 + 4i)\bar{z} - 92 = 0$ … (1)

$(1 - i)z + (1 + i)\bar{z} - 16 = 0$ … (2)

Let $z = x + iy \Rightarrow \bar{z} = x - iy$

$\therefore z + \bar{z} = 2x,\ z - \bar{z} = 2iy$ … (3)

From (1),

$|z|^2 + 3(z + \bar{z}) - 4i(z - \bar{z}) - 92 = 0$

$\Rightarrow x^2 + y^2 + 6x + 8y - 92 = 0$ … (4)

From (2),

$(z + \bar{z}) - i(z - \bar{z}) - 16 = 0$

$\Rightarrow 2x + 2y - 16 = 0$

$\Rightarrow x + y = 8$ … (5)

$\Rightarrow y = (8 - x)$

$\therefore$ From () and (5)

$x^2 + (8 - x)^2 + 6x + 8(8 - x) - 92 = 0$

$\Rightarrow x^2 + 64 - 16x + 6x + 64 - 8x - 92 = 0$

$\Rightarrow 2x^2 - 18x + 36 = 0 \Rightarrow x^2 - 9x + 18 = 0$

$\Rightarrow (x - 6)(x - 3) = 0 \Rightarrow x = 3, 6.$

$\therefore$ From (5),

$y = 5, 2$

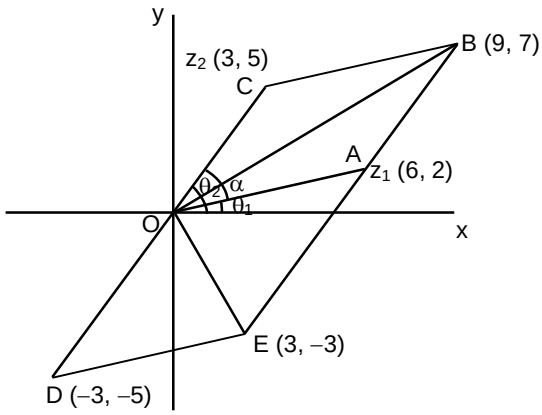
$\therefore z_2 = 3 + 5i = (3, 5)$ and $z_1 = (6 + 2i) = (6, 2)$

$z_1 + z_2 = (3 + 5i) + (6 + 2i) = (9 + 7i) = (9, 7)$

$(z_1 - z_2) = (6 + 2i) - (3 + 5i) = (3 - 3i)$

$= (3, -3).$

Clearly, quadrilaterals OABC and ODEA are the parallelogram of same area.



$\tan\theta_1 = \frac{z}{6} = \frac{1}{3}$, $\tan\theta_2 = \frac{5}{3}$

$\therefore \alpha = (\theta_2 - \theta_1) \Rightarrow \tan\alpha = \left(\frac{\frac{5}{3} - \frac{1}{3}}{1 - \frac{5}{3} \times \frac{1}{3}}\right)$

$\Rightarrow \tan\alpha = \frac{4/3}{4/3} = 1$

$\Rightarrow \alpha = 45°.$

Total area = 2 area of parallelogram OABC

$= = 2|z_1||z_2| \sin\alpha$

$= 2 \times \sqrt{40} \times \sqrt{34} \times \frac{1}{\sqrt{2}}$

$= 2\sqrt{20} \times \sqrt{34} = 4\sqrt{170}$

$= 4 \times 13.04 = 52.16$ sq. units.

53. 00003.07

Sol. Let $y = \frac{3x^2 + 2x - 1}{x^2 + x + 1}$

$\Rightarrow (y - 3)x^2 + (y - 2)x + y + 1 = 0.$

As $x \in R$, $D \geq 0 \Rightarrow (y - 2)^2 - 4(y - 3)(y + 1) \geq 0$

$\Rightarrow 3y^2 - 4y - 16 \leq 0.$

Clearly y lies between the roots.

Roots of the equation $3y^2 - 4y - 16 = 0$ are

$y = \frac{4 \pm \sqrt{208}}{6} = \frac{2 \pm \sqrt{52}}{3}$.

$\therefore$ absolute maximum of the expression = $\frac{2+\sqrt{52}}{3} = \alpha$.

54. 00011.00

Sol. $CT_1 = CT_2 = x$

AB = 20

AC = 11

BC = 13

In ΔABC $\sin A = \frac{3}{5}$ and $\sin B = \frac{33}{65}$ also

$\frac{r}{\sin A} + \frac{r}{\sin B} = AB \Rightarrow r = \frac{11}{2}$.

$\Rightarrow [\alpha] = 3$.